

Lesson 13 Distributed Communication

Instruction

1. Introduction

Multicomputer communication refers to the process of information transmission and exchange between multiple computers or nodes in a network.

The ROS-based intermediate piece is DDS. In the same network, the distributed communication can be implemented by using the domain DDS's ID mechanics (`ROS_DOMAIN_ID`). The general progress: before starting nodes, the value of domain ID can be set. If different nodes have the same domain ID, they can freely discover and communicate with each other. Conversely, if the domain ID values are different, communication cannot be established. By default, all nodes use a domain ID of 0 at startup. In other words, as long as you ensure that they are in the same network, no configuration is required, and nodes on different ROS2 devices can achieve distributed communication.

The distributed communication has extensive application scenarios such as autonomous vehicle formations, drone formations, remote control, and so on. The interaction of these data relies on distributed communication.

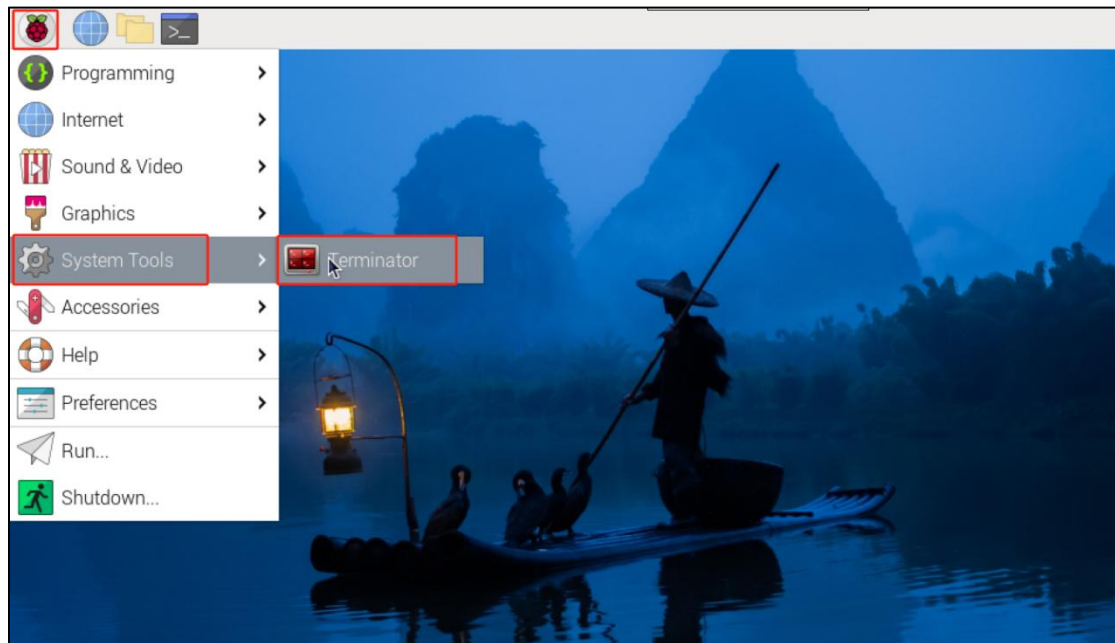
2. ROS2 Distributed Network Group

ROS2 provides a mechanism called DOMAIN, similar to grouping, where terminals within the same DOMAIN can communicate.

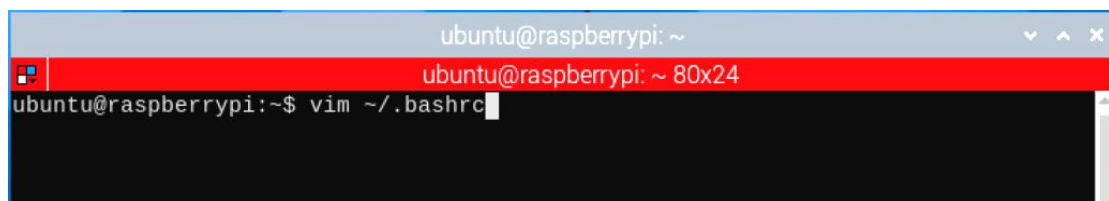
By default, all ROS2 nodes uses a domain ID of 0. To avoid messages confusion, devices from different groups running ROS2 within the same

network should use different domain ID. It is recommended to use ID between 0-101.

- 1) Click  and select “**System Tools ->Terminator**” in sequence.



- 2) Enter the command “**vim ~/.bashrc**” to open the bashrc file using VIM text editor.



- 3) Press “i” to enter the insert mode, and enter “**export ROS_DOMAIN_ID=25**” in corresponding location. After you’ve finishing modifications, press “Esc” and enter “:wq” to save and exit.

```
. /usr/share/bash-completion/bash_completion
elif [ -f /etc/bash_completion ]; then
. /etc/bash_completion
fi
fi
source /opt/ros/humble/setup.bash
source ~/hiwonder_ws/install/setup.bash
export ROS_DOMAIN_ID=25
-- INSERT --
```

4) Enter the command “**source ~/.bashrc**” to make the environment variables make effect.

```
ubuntu@raspberrypi:~$ source ~/.bashrc
ubuntu@raspberrypi:~$
```

Add the statement “**export ROS_DOMAIN_ID=25**” to the barshrc files of other terminals to assign them to the same group (i.e., domain ID 25), enabling distributed communication. If different domain IDs are assigned, communication between them will not be possible.

3. Comparison of ROS1 and ROS in Distributed Communication Mechanisms.

The distributed Communication Mechanism of ROS1:

- ◆ Utilizes anonymous TCP/IP-based communication for distributed communication.
- ◆ Communication adopts publish/subscribe mode through central master
- ◆ Nodes need to actively register with the master to obtain information about peer nodes.
- ◆ The reliability of communication depends on the central node master.
- ◆

The distributed communication mechanism of ROS2:

- ◆ Utilizing the common transmission protocols such as TCP, UDP, DDS, etc.
- ◆ Introduce distributed service discovery with mechanisms for discovery, authentication and authorization mechanism.
- ◆ Nodes communicate directly with peer nodes without the need for a central mater.
- ◆ ZeroMQ is deployed by default to build communication middleware, enabling decentralized distributed communication.
- ◆ Support both point-to-point and multicast communication, providing better scalability.
- ◆ Stronger fault tolerance, ensuring communication can be maintained even if nodes go offline.